

Andrew Li

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EDUCATION

Georgia Institute of Technology

Atlanta, GA

B.S./M.S. in Computer Science, Undergraduate GPA: 3.74

Aug. 2023 – May 2027

- **Coursework:** Machine Learning, Computer Vision, Perception and Robotics, Intro to AI, Linear Algebra, DSA

EXPERIENCE

Lunar Lab @ Georgia Tech

Atlanta, GA

Research Assistant

Apr. 2026 – Present

- Engineering a real2sim pipeline turning a single LiDAR and tri-camera capture into a photorealistic, collision-ready digital twin, implementing SOTA research in SLAM, 3D mesh reconstruction, 3DGS, and asset generation
- Optimized joint mesh + Gaussian-splatting reconstruction of large-scale indoor scenes, improving fine-detail surface coverage by 14%, currently lifting novel-view PSNR by 1.3 dB (19.4 to 20.7) and SSIM from 0.71 to 0.75
- Generating physics-aware manipulable assets from a single capture with YOLO11x-seg + SAM2, feeding image-to-URDF/MJCF generation, eliminating manual simulation asset authoring for downstream tasks

Sandia National Laboratories

Albuquerque, NM (On-site → Remote)

Year-Round AI for Autonomy Intern

May 2025 – Present

- Developing modular C++ firmware for a platform-agnostic microcontroller-based drone flight logging module to ensure high-frequency capture of inertial, barometric, and GPS data
- Engineered a semantic navigation stack in simulation for a Spot robot dog, utilizing Gemma 3 locally via Ollama for contextual reasoning and behavior trees to orchestrate high-level planning
- Developed a custom C++ ROS2 interface for Kromek GR1+ spectrometers and SparkFun GPS, leading Jackal UGV field trials for real-time radiation mapping as detailed in report [SAND2025-12871R](#)

Humanoid Robotics @ Georgia Tech

Atlanta, GA

Software Lead

Jan. 2026 – Present

- Leading a team of 5 developers implementing PPO neural network policies for dynamic bipedal locomotion, balance recovery, and soccer dribbling, with a standardized Docker-based Ubuntu workflow for cross-platform reproducibility

Georgia Tech RoboWrestling

Atlanta, GA

Software Mentor

Sep. 2023 – Feb. 2025

- Developed embedded C++ firmware on Teensy 4.1 microcontrollers for autonomous opponent detection
- Mentored 15+ students on 500g robot sumo teams from design to competition

Boston University Morphable Biorobotics Lab

Boston, MA

Research Assistant

Jun. 2022 – Aug. 2022

- Designed a soft-robotic pick-and-place manipulator and prototyped minimally invasive cardiac devices

PROJECTS

Basketball Rebounding Robot

Sep. 2025 – Present

- Leveraged ZED VSLAM and OpenCV on a Jetson Orin Nano for autonomous navigation
- Created a hierarchical state machine in ROS2 to coordinate autonomous sub-systems, including target tracking, ball acquisition, and launching sequences
- Engineered a custom flywheel system and x-drive chassis using Fusion 360, including all fabrication and assembly

TECHNICAL SKILLS

Languages: Python, C++, Java

AI/Robotics/Sim: Isaac Sim/Lab/Gym, MuJoCo, PPO, 3DGS, SLAM, Behavior Trees, ROS2, Rviz, Computer Vision

Hardware: Jetson Orin Nano, Boston Dynamics Spot, Teensy 4.1, Arduino, Jackal UGV, LiDAR, ZED stereo camera

Tools: Docker, Git, Linux (Ubuntu), Fusion 360